Universität Bern - Computer Graphics Group

Final Exam

Applied Optimization

Notes and Rules:

- You have to work on task 1 (Declaration of Independent Work) and scan your student ID together with the exam. Otherwise your submission is invalid and will be graded as 1.
- Tasks marked as bonus are harder than the rest. We recommend solving the regular ones first.
- You do not necessarily need to get all 100 points to obtain a 6.
- The only aid allowed is an A4 sheet of double-sided, handwritten notes. In particular, electronic devices may only be used to view the exam PDF, participate in the Zoom session and for scanning the exam.
- Position your webcam so that you are clearly and visibly doing the exam. Note that your work should not be legible by your fellow students.
- Use a new sheet of paper for each question. You can use more than one sheet for a task. Leave enough space between subtasks for your own corrections and for clarity.
- The exam can only be taken on paper. Write legibly with a permanent blue or black pen. We will only grade answers that we can clearly read.
- Use white paper for your answers.
- Each sheet must begin with a header with the following content:
  - Page number
  - Task Number
  - Name & matriculation number
  - Signature
- You must answer all questions in English.
- Submission is done by uploading scans of your solution on ILIAS. If you experience technical problems and are unable to do so, you can also send your solution by e-mail from your campus email account to `opt-staff@list.inf.unibe.ch` Keep the original of your exam safe.
- You can work on the exam up to the deadline. Scanning your solution will come after that.

- When scanning, merge all pages into a single PDF, in the correct order. If you don't have enough time to do so, uploading files individually is fine.

- Please do not directly ask your questions on Zoom. If you have any questions or issues, you can notify us with the Zoom chat or with the "show of hands" button. We can switch to a one-on-one meeting if needed.

- Contact us immediately if you have any technical problems. Via Zoom chat, e-mail to `opt-staff@list.inf.unibe.ch` or phone number +41 76 360 62 55.

The student is deemed to have failed the course if they are found cheating or violating any of the above regulations.

Tasks and points summary

Exercise	Points	*Earned points*
1. Declaration of Independent Work	0	
2. Convex Sets	10	
3. Convex Functions	12	
4. Line Search	16	
5. Trust Region Method	22	
6. Active Set Method	16	
7. Algorithms	16	
8. Programming	8	
Total	100	

1 Declaration of Independent Work (0 point)

Read the following *Declaration of Independent Work:*

I declare herewith that I wrote this thesis/examination on my own, without the help of others and without using any unpermitted sources of information (internet, books, etc.). Wherever I have used permitted sources of information, I have made this explicitly clear within my text and I have listed the referenced sources. I understand that if I do not follow these rules my examination will be assigned the mark 1.0, and that the Senate of the University of Bern is authorized to revoke the title awarded on the basis of this examination (according to Article 36, paragraph 1, litera r of the University Act of September 5th, 1996).

For the purposes of evaluation and verification of compliance with the declaration of originality and the regulations governing plagiarism, I hereby grant the University of Bern the right to process my personal data and to perform the acts of use this requires, in particular, to reproduce the written thesis and to store it permanently in a database, and to use said database, or to make said database available, to enable comparison with theses submitted by others.

1.1. **0 point** Write down the following sentence on your solution and sign it:
"By submitting this exam, I confirm that I have the read and understood the *Declaration of Independent Work*, that I have taken note of the rules of examination and that I accept them."

1.2. **0 point** Put your Student ID on the same page when you scan it.

WARNING: Both parts of this task are mandatory.
Failure to complete either of them will result in a failed course.

2 Convex Sets (10 points)

Sketch the following sets and indicate which are convex:
Note: make sure to chose a relevant interval for each sketch.

2.1. $S_A = \{x \in \mathbb{R}^2 \,|\, 1 \leq ||x||_2 \leq 2\}$

2.2. $S_B = \{x \in \mathbb{R}^2 \,|\, Ax \succcurlyeq b\}$, with $A = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$

2.3. $S_C = \{(x,y) \in \mathbb{R} \times [-1,1] \,|\, x = y^2\}$

2.4. $S_D = \{x \in \mathbb{R} \,|\, 0 \leq ||x||_1 \leq \pi,\, |cos(x)| \leq 1/2\}$

3 Convex Functions and Perspective (12 points)

Let us define the *perspective* $g : \mathbb{R}^{n+1} \to \mathbb{R}$ of $f : \mathbb{R}^n \to \mathbb{R}$ as

$$g(x,t) = t\,f(\frac{x}{t}),\ \mathrm{dom}\, g = \{(x,t) \mid \frac{x}{t} \in \mathrm{dom} f,\ t > 0\}$$

3.1. **(4 points)** Show that the perspective of the function $f : \mathbb{R}^+ \to \mathbb{R}$, $f(x) = -\log(x)$ is convex.
NOTE: You **cannot** use your answer to 3.2 to solve 3.1.

3.2. **(8 points)** Show that the perspective of any convex function is convex.

4 Line Search (16 points)

Consider the quadratic objective function

$$f(x) = \frac{1}{2}x^T Hx + q^T x$$

with

$$H = \frac{1}{2}\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

and

$$q = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

4.1. **(6 points)** Given a starting point $x_0 = (0,1)^T$, find the optimal point x^* in the search direction $\Delta x_0 = (-2,0)^T$. Justify your answer by showing all intermediary steps.

4.2. **(7 points)** Can we use Newton's method to find a valid descent direction Δ_x for this problem, starting from x_0? Justify your answer with all necessary computations.

4.3. **(3 points)** Express a matrix M such that

$$\hat{f}(x) = \frac{1}{2}x^T(H+M)x + g^T x$$

is a convex problem.

BONUS **(2 points)** Is the global minimum of the modified, convex problem $\hat{f}(x)$ also a local or global minimum of the original problem? Briefly justify your answer.

5 Trust-region method (22 points)

Let us consider the minimization of the following function

$$f(x,y) = \cos(100x)\exp\left[\sin(y)\right]$$

This function being highly anisotropic, the minimization will be done by using an **exact** trust-region method.

The trust-region is based on the following model function

$$m(v) = f(x,y) + v^T \cdot \nabla f_{(x,y)} + \frac{1}{2} v^T \cdot \nabla^2 f_{(x,y)} \cdot v$$

with this constraint

$$||v||_2 \leq 10^{-2}$$

5.1. **(6 points)** Explicitly express the corresponding trust-region problem at $(x_0, y_0) = (0,0)^T$.

5.2. **(4 points)** State the KKT conditions yielding a global optimum v^* for this specific problem.

5.3. **(6 points)** Compute exactly the global optimum v^*.

5.4. **(4 points)** Compute the ratio between the **actual** and **predicted** reductions.
Hint: $\cos(\theta) \approx 1 - \frac{1}{2}\theta^2$, $\sin(\theta) \approx \theta$ for small $|\theta|$ and $\exp(z) \approx 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3$

5.5. **(2 points)** Based on your answer to **8.4**, how would you adapt the trust-region's size ?

6 Active Set Method (16 points)

Consider the following optimization problem with $x, y \in \mathbb{R}$

$$
\begin{aligned}
\text{minimize} \quad & f(x,y) = \frac{1}{2}(x^2 + y^2) - 4x - 4y \\
\text{subject to:} \quad & -x \leq 0 && (C_1) \\
& -y \leq 0 && (C_2) \\
& -5 + x + y \leq 0 && (C_3) \\
& 2 - x - y \leq 0 && (C_4)
\end{aligned}
$$

6.1. **(4 point)** Sketch the feasible set below, give the minimum x^* of the unconstrained problem and indicate x^* on your sketch.

6.2. **(6 points)** Sketch the optimization path P_1 of the Active Set Method for convex QP as discussed in the lecture, starting from $(x_0, y_0) = (3, 0)$ with initial active set $A_0 = \{C_2\}$ on the same drawing. Draw a $\times$ at each discrete point of P_1.

6.3. **(6 points)** Sketch the optimization path P_2 of the Active Set Method for convex QP as discussed in the lecture, starting from $(x_0, y_0) = (0, 2)$ with initial active set $A_0 = \{C_4\}$ on the same drawing. Draw a $\circ$ at each discrete point of P_2.

7 Algorithms (16 points)

Consider the task of optimizing a problem $f(x)$, $x \in \mathbb{R}^n$, $n = 10^6$ with $f \in C^d$ (i.e. d times differentiable), with m independent linear equality constraints and with or without a feasible starting point x_0. The following algorithms are available, as implemented in the lecture:

(a) L-BFGS

(b) Feasible start Newton's method with linear equality constraints

(c) Quadratic penalty method with Gradient Descent

(d) Active-Set method

(e) Big M method

(f) Sequential Quadratic Programming

Note: algorithms (e), (f) and (g) all use feasible start Newton's method with linear equality constraints, Hessian projection as their underlying optimization method.

7.1. **(12 points)** For each set of parameters given in the table below, give the list of suitable methods to obtain a (local) minimum.
Note: You will get -1 point every time you give one algorithm that is actually not suitable (your total for 7.1 cannot go below 0 points).

	f convex	sparse	m	C^d	feasible x_0	suitable methods
1	yes	yes	10^5	C^∞	*yes*	
2	no	no	10^5	C^∞	*no*	
3	no	yes	0	C^1	*yes*	
4	yes	yes	10	C^1	*yes*	

7.2. **(4 points)** For each problem, indicate which method you would use to be most efficient and ***briefly*** justify why.

BONUS **(2 points)** Which algorithm(s) would you use to find the global optimum for a non-convex, dense problem with $n = 10^9$, $m = 10^9$, $f \in C^\infty$ and a feasible starting point x_0 ? Justify your answer.

8 Programming (8 points)

We have the following code to solve optimization problems with Newton's method. The code works but runs very slowly for sparse problems with a very large number of variables.

```
using Vec = Eigen::VectorXd<double>;
using Mat = Eigen::MatrixXd<double>;

Vec solve_with_newton_method(ObjectiveFunction *_obj_func,
                             const Vec& _initial_x,
                             const double _eps,
                             const int _max_iters) {


    int n = _obj_func->n_unknowns();
    Vec x(_initial_x), g(n), delta_x(n);

    double fp = std::numeric_limits<double>::max();

    int iter(0);
    do {
        ++iter;

        _obj_func->eval_gradient(x, g);
        _obj_func->eval_hessian(x, H);

        g = -H.inverse() * delta_x;

        double lambda2 = -g.transpose() * delta_x;
        double f = _obj_func->eval_f(x);

        if (lambda2 <= (_eps * _eps) || fp <= f) break;

        double t = LineSearch::backtracking_line_search(_obj_func, x, g, delta_x, 1.0);

        x += t * delta_x;
        fp = f;
    } while (iter < _max_iters);

    return x;
}
```

8.1. **(3 points)** Identify what part or parts of the algorithm is making the optimization of large sparse problems so slow.

8.2. **(5 points)** Describe how to significantly improve performance.